

Student Activity: Comprehensive Software Project Management Exercise

Objective:

Students will gain hands-on experience in software project management by exploring and applying key concepts such as SPM tools, PMI's Knowledge Areas, Lifecycle Relationships, and PMI Process Groups. They will also learn how to identify and avoid classic mistakes in project management and develop essential project documents like a Project Charter and Statement of Work.

Activity Overview:

This activity is divided into several tasks, each designed to cover different aspects of Software Project Management (SPM). Students will work in groups to simulate a project management environment, where they will plan, execute, and monitor a hypothetical software project using SPM tools and methodologies.

Materials Needed:

- Access to SPM tools (e.g., MS Project, Trello, JIRA)
 - PMI's PMBOK Guide (Project Management Body of Knowledge) or relevant resources
 - Computer with internet access
 - Project template documents (Project Charter, Statement of Work)
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Activity Breakdown

Task 1: Introduction to SPM Tools

- **Objective:** Familiarize students with various SPM tools and their functionalities.
- **Instructions:**
 1. Divide students into small groups and assign each group a different SPM tool (e.g., MS Project, Trello, JIRA).
 2. Each group will explore their assigned tool, focusing on its features such as task management, resource allocation, and tracking.
 3. Groups will create a short presentation or demonstration on how to use the tool for project management tasks.

Task 2: PMI's Knowledge Areas and Process Groups

- **Objective:** Understand and apply PMI's Knowledge Areas and Process Groups in a simulated project.
- **Instructions:**
 1. Each group will be assigned a specific Knowledge Area (e.g., Scope Management, Time Management) and a Process Group (e.g., Initiating, Planning, Executing).

2. Groups will discuss how their assigned Knowledge Area and Process Group are applied in managing a project.
3. Groups will prepare a short report or presentation on their findings, including examples from the hypothetical project.

Task 3: Lifecycle Relationships and Technical Fundamentals in SPM

- **Objective:** Explore the relationships between different lifecycle stages and understand the technical fundamentals in SPM.
- **Instructions:**
 1. Students will map out the lifecycle of a software project, identifying key stages such as initiation, planning, execution, monitoring, and closing.
 2. They will discuss the technical fundamentals involved at each stage, such as requirements gathering, coding, testing, and deployment.
 3. Each group will create a lifecycle diagram and explain the dependencies and relationships between different stages.

Task 4: Identifying and Avoiding Classic Mistakes

- **Objective:** Recognize common pitfalls in project management and learn strategies to avoid them.
- **Instructions:**
 1. Students will research classic mistakes in project management related to product, process, people, and technology.
 2. Each group will identify one or two common mistakes and discuss their impact on project success.
 3. Groups will present their findings and suggest strategies to avoid these mistakes in future projects.

Task 5: Developing a Project Charter and Statement of Work

- **Objective:** Create essential project documentation to outline the project's scope, objectives, and deliverables.
- **Instructions:**
 1. Each group will draft a Project Charter for the hypothetical project, including the project's purpose, objectives, and stakeholders.
 2. Groups will then develop a Statement of Work (SOW) that details the project's scope, deliverables, timelines, and responsibilities.
 3. Students will review and provide feedback on each other's documents to ensure clarity and completeness.

Task 6: Applying PMI Process Groups

- **Objective:** Simulate the management of a project by applying PMI's Process Groups.
- **Instructions:**

1. Each group will be responsible for managing a different phase of the project, corresponding to one of the PMI Process Groups (Initiating, Planning, Executing, Monitoring & Controlling, Closing).
 2. Students will document the activities, decisions, and challenges encountered during their phase.
 3. Groups will collaborate to ensure a smooth transition between phases, mimicking real-world project management scenarios.
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Assessment Criteria:

- **Participation and Collaboration:** Active involvement in group discussions and activities.
- **Understanding of SPM Tools:** Ability to effectively use the assigned SPM tool for project management tasks.
- **Application of PMI Knowledge Areas and Process Groups:** Accurate and thorough application of PMI concepts in the simulated project.
- **Quality of Project Documents:** Completeness and professionalism in the Project Charter and Statement of Work.
- **Identification of Classic Mistakes:** Insightfulness in recognizing common pitfalls and suggesting practical solutions.
- **Presentation and Reporting:** Clarity and effectiveness in presenting findings and project progress.